

FACULTÉ DES SCIENCES ET TECHNOLOGIES (FST)

Administration Des Systèmes Informatiques

TD1

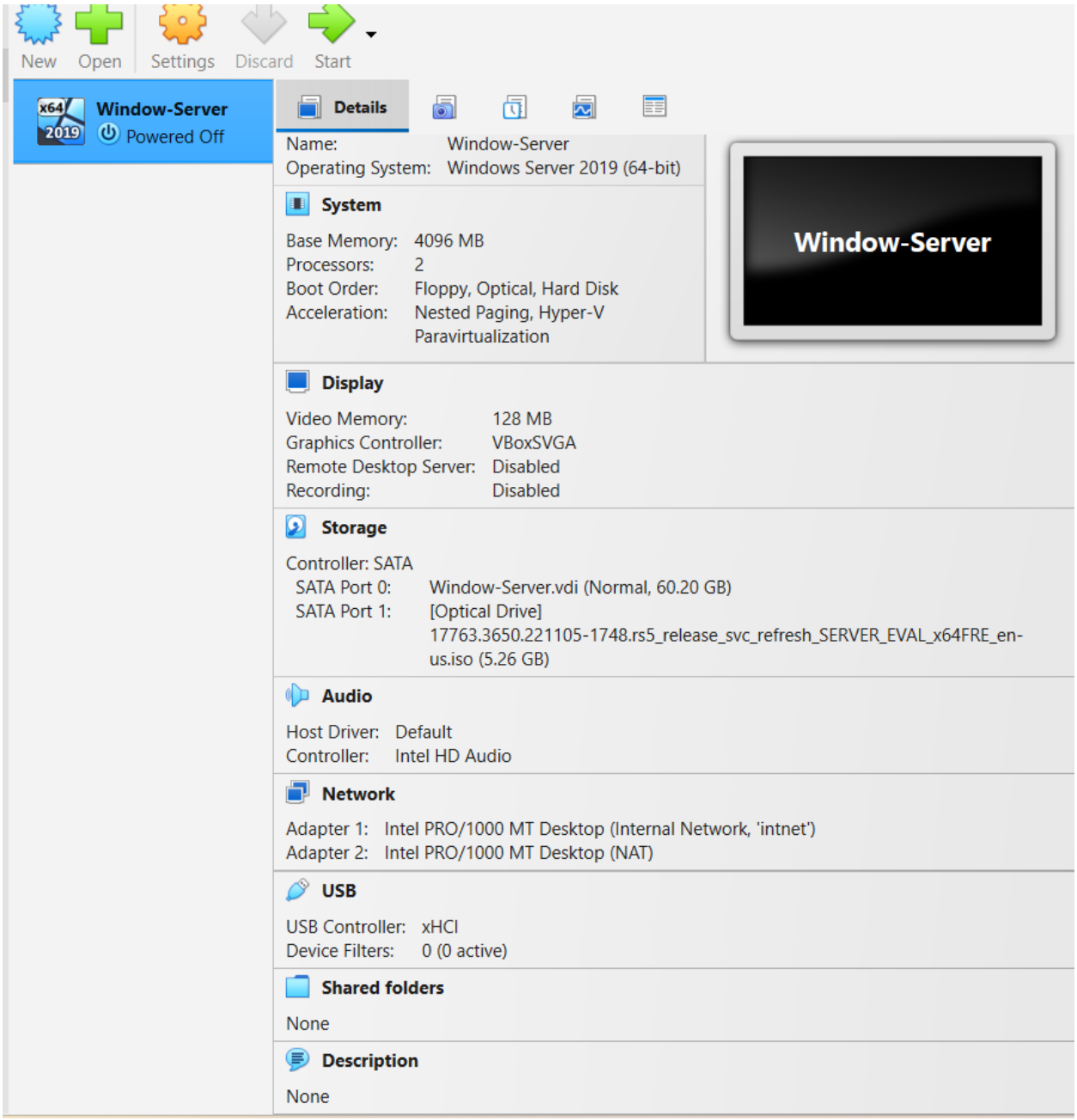
Niveau : L4

Préparé : **Donsam Jean Gabard NOEL**

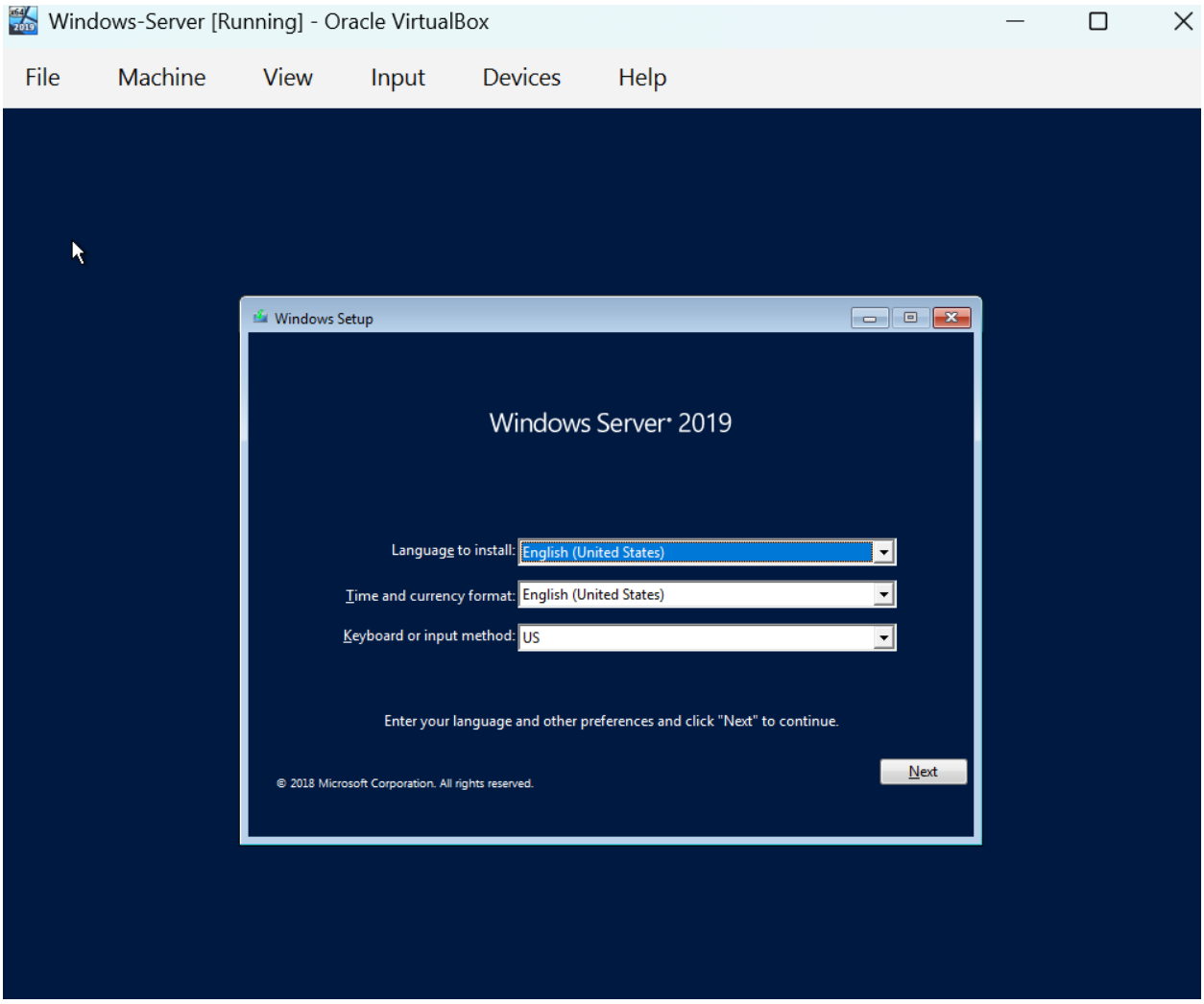
Sous la direction de: **M. Ismaël SAINT-AMOUR**

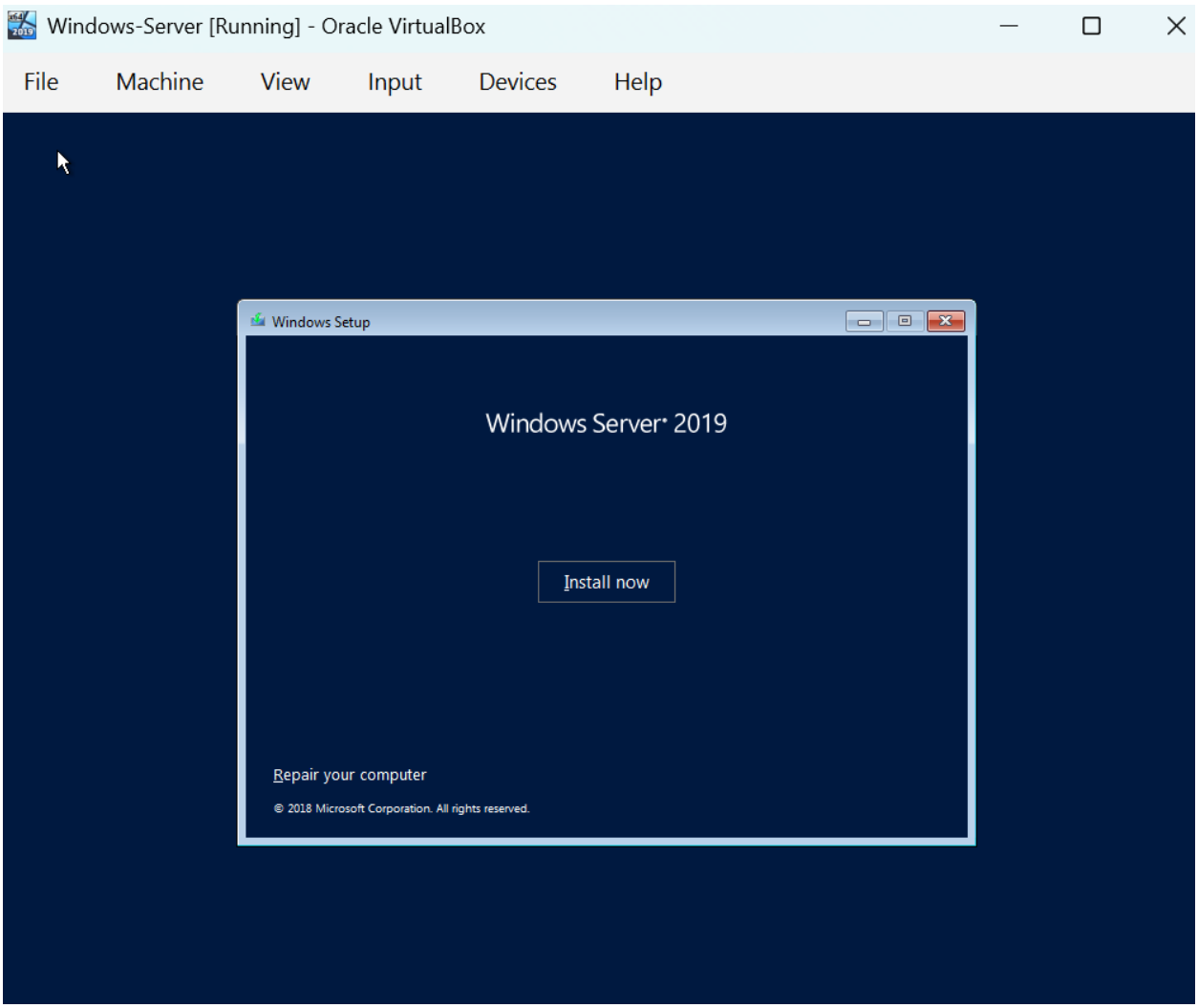
Le 24/10/2026

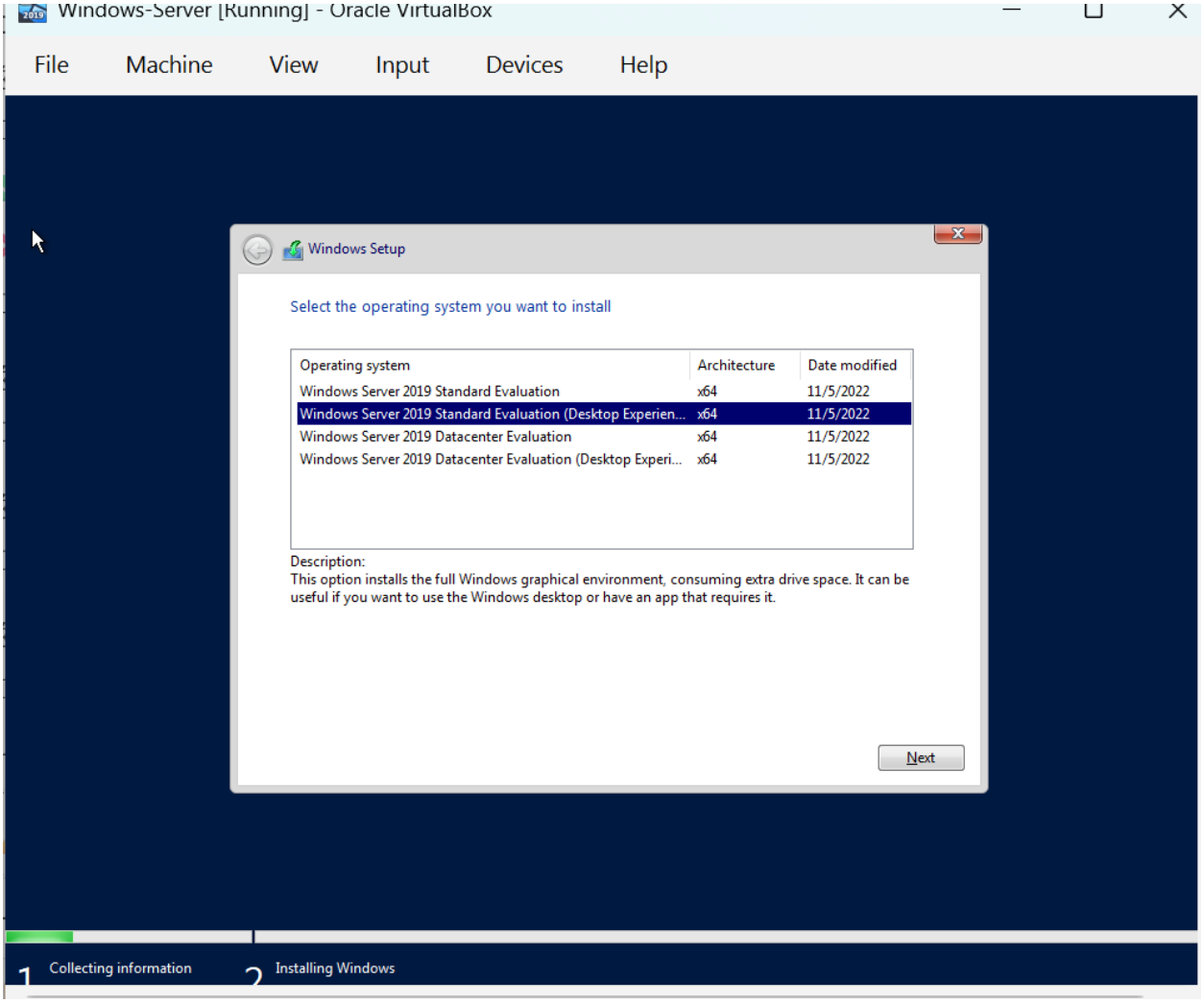
1-Configuration du VM Windows-Server

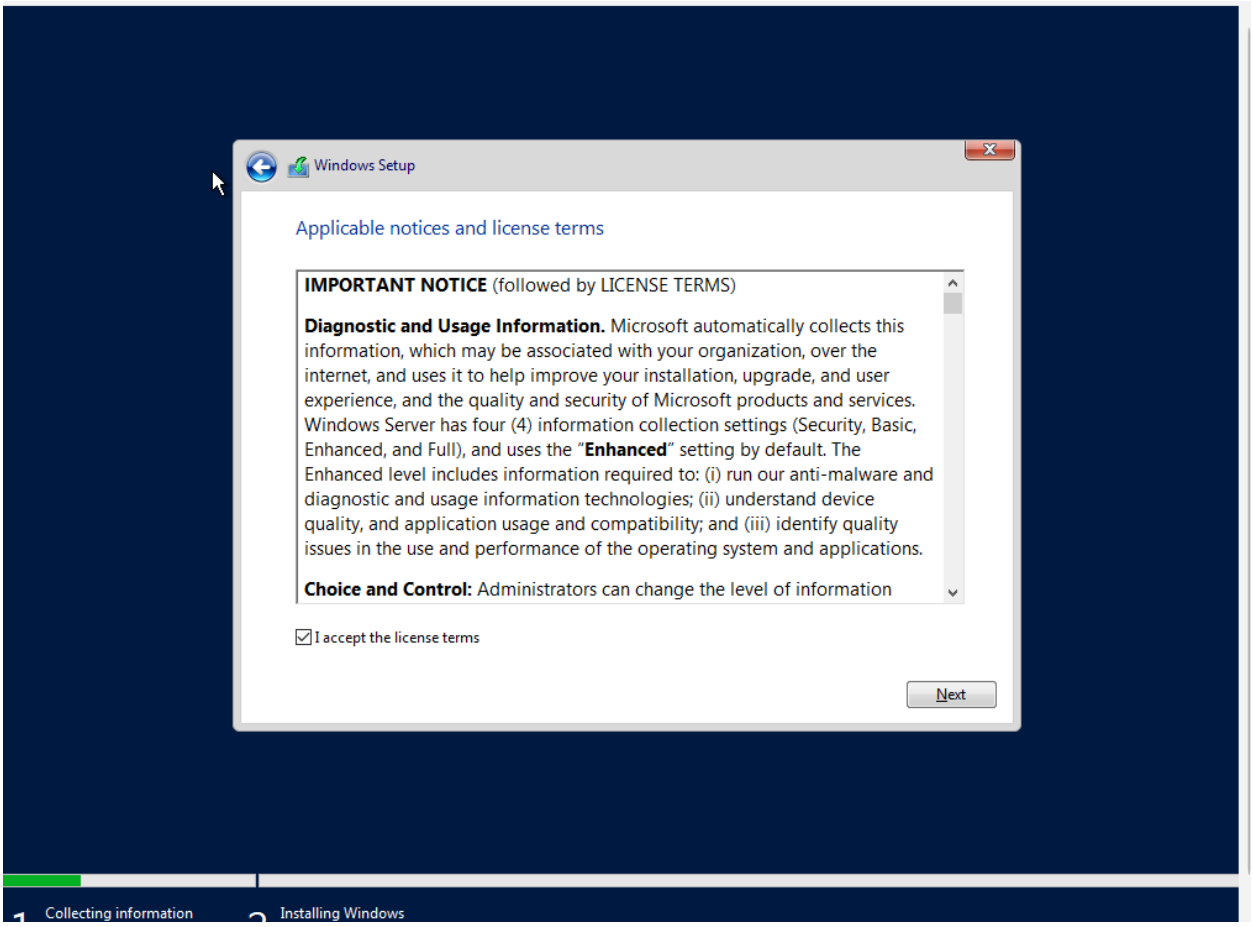


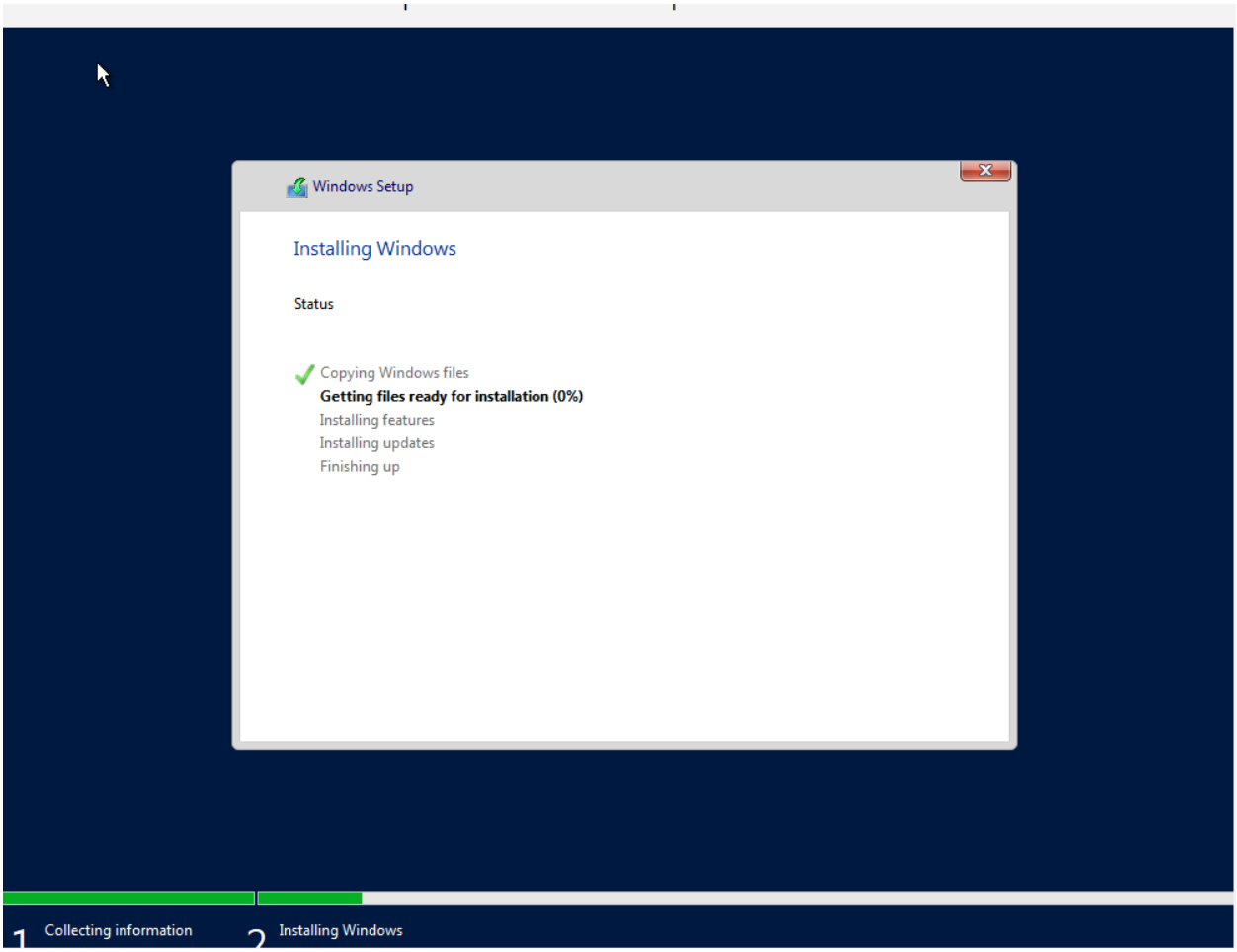
2-Installation de Windows Server 2019

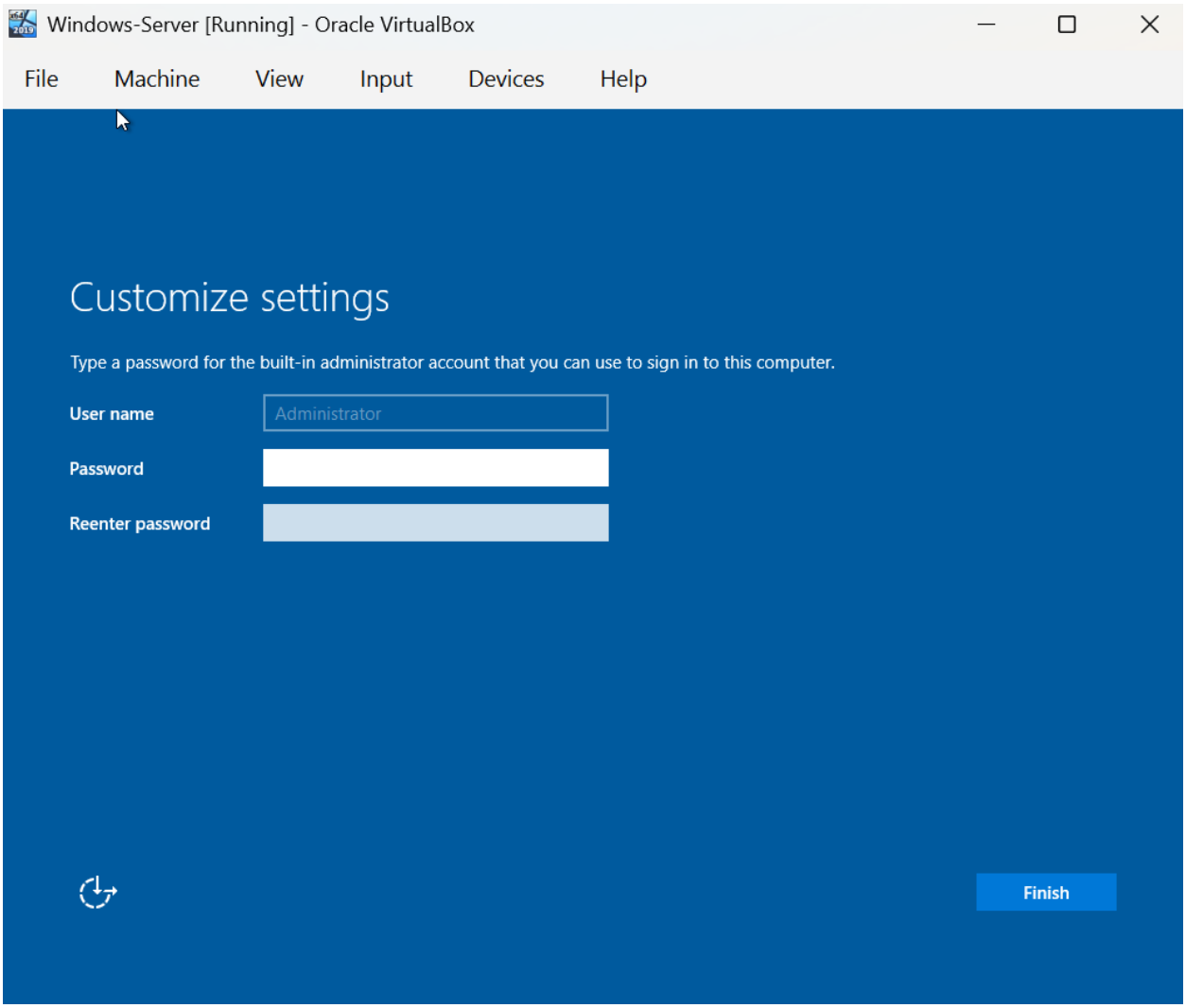


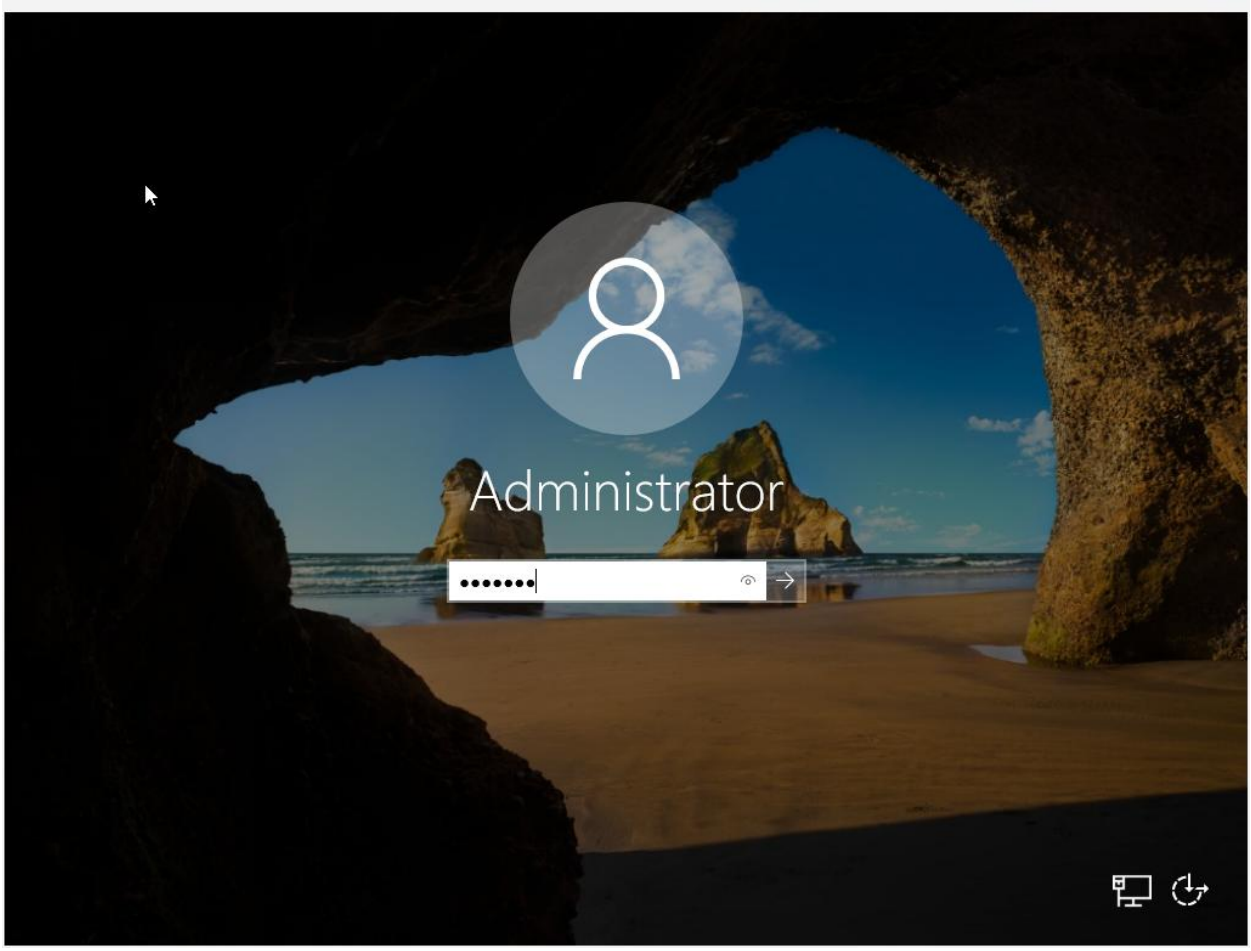

Windows-Server [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Windows Setup
Windows Server 2019
Install now
Repair your computer
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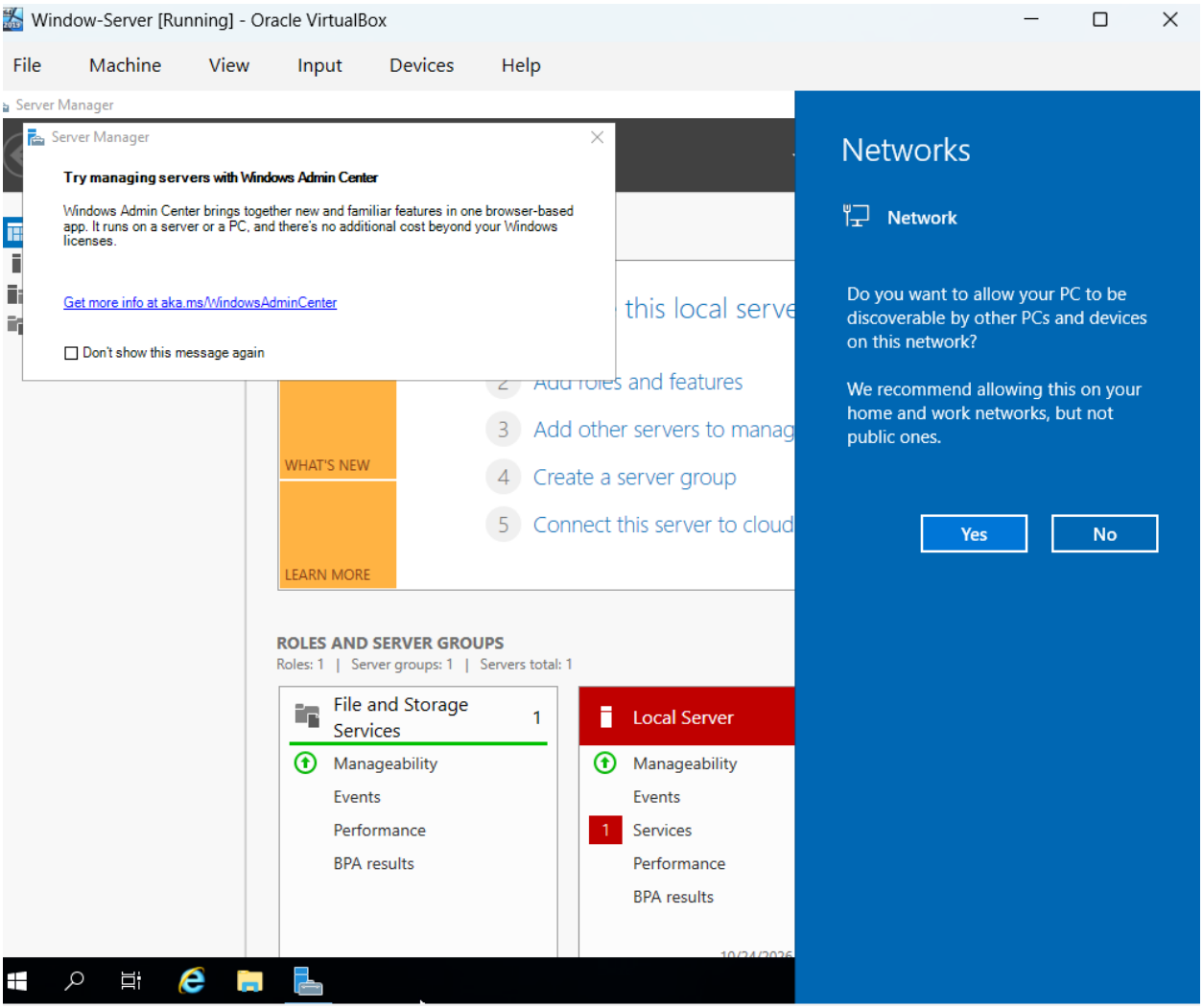

Windows-Server [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Windows Setup
Select the operating system you want to install
Operating system
Architecture
Date modified
Windows Server 2019 Standard Evaluation
x64
11/5/2022
Windows Server 2019 Standard Evaluation (Desktop Experien...
x64
11/5/2022
Windows Server 2019 Datacenter Evaluation
x64
11/5/2022
Windows Server 2019 Datacenter Evaluation (Desktop Experi...
x64
11/5/2022
Description:
This option installs the full Windows graphical environment, consuming extra drive space. It can be useful if you want to use the Windows desktop or have an app that requires it.
Next
1 Collecting information
2 Installing Windows


Windows Setup
Applicable notices and license terms
IMPORTANT NOTICE (followed by LICENSE TERMS)
Diagnostic and Usage Information. Microsoft automatically collects this information, which may be associated with your organization, over the internet, and uses it to help improve your installation, upgrade, and user experience, and the quality and security of Microsoft products and services. Windows Server has four (4) information collection settings (Security, Basic, Enhanced, and Full), and uses the "Enhanced" setting by default. The Enhanced level includes information required to: (i) run our anti-malware and diagnostic and usage information technologies; (ii) understand device quality, and application usage and compatibility; and (iii) identify quality issues in the use and performance of the operating system and applications.
Choice and Control: Administrators can change the level of information
I accept the license terms
Next
Collecting information
Installing Windows


Windows Setup
Installing Windows
Status
Copying Windows files
Getting files ready for installation (0%)
Installing features
Installing updates
Finishing up
1 Collecting information
2 Installing Windows


Windows-Server [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Customize settings
Type a password for the built-in administrator account that you can use to sign in to this computer.
User name
Administrator
Password
Reenter password
Finish


Administrator
•••••••



1. Quelle différence existe-t-il entre Server Core et l'installation avec Expérience utilisateur ?

R-Server Core n'a pas d'interface graphique : on l'administre en ligne de commande.Tandis qu' Expérience utilisateur a le Bureau et les outils graphiques, c'est plus simple à utiliser.

2. Pourquoi utiliser une machine virtuelle pour ce TD ?

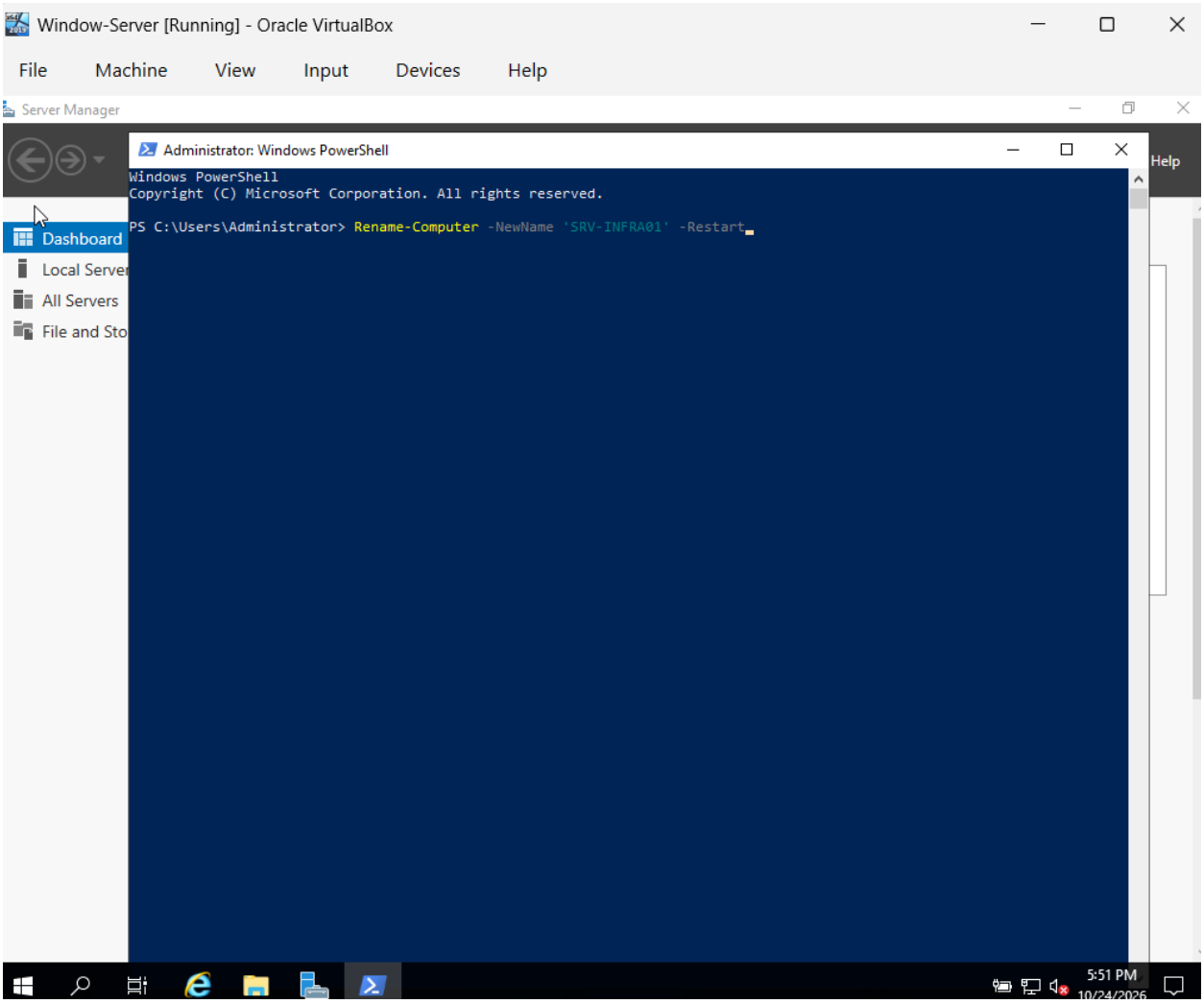
R-Pour tester sans risque pour le vrai PC, faire tourner un serveur et un client sur un seul ordinateur, et pouvoir revenir en arrière en cas d'erreur.

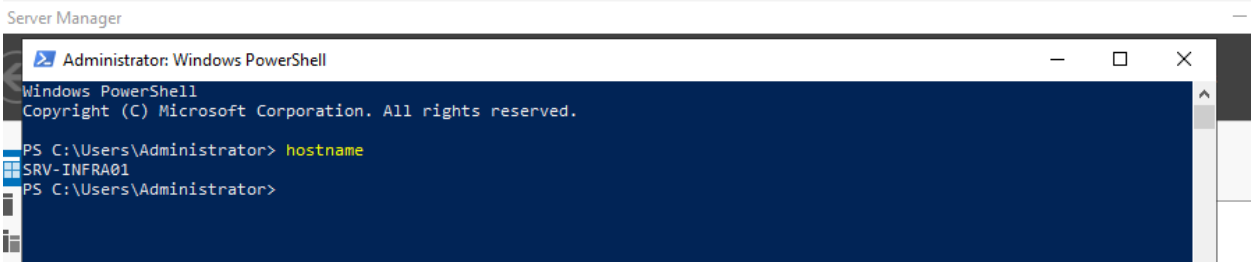
3. Quel est le rôle du compte Administrateur ?

R-C'est le compte qui a tous les droits : il installe les rôles, configure le réseau, gère les utilisateurs et les autorisations.

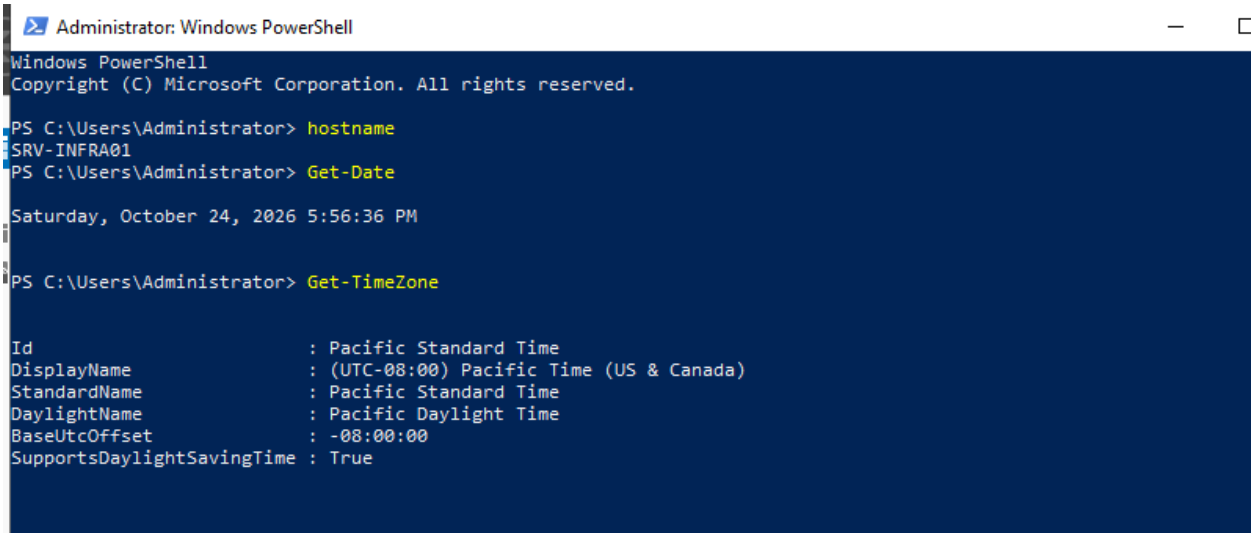
3-Configuration initiale du serveur

A-Renommer le serveur

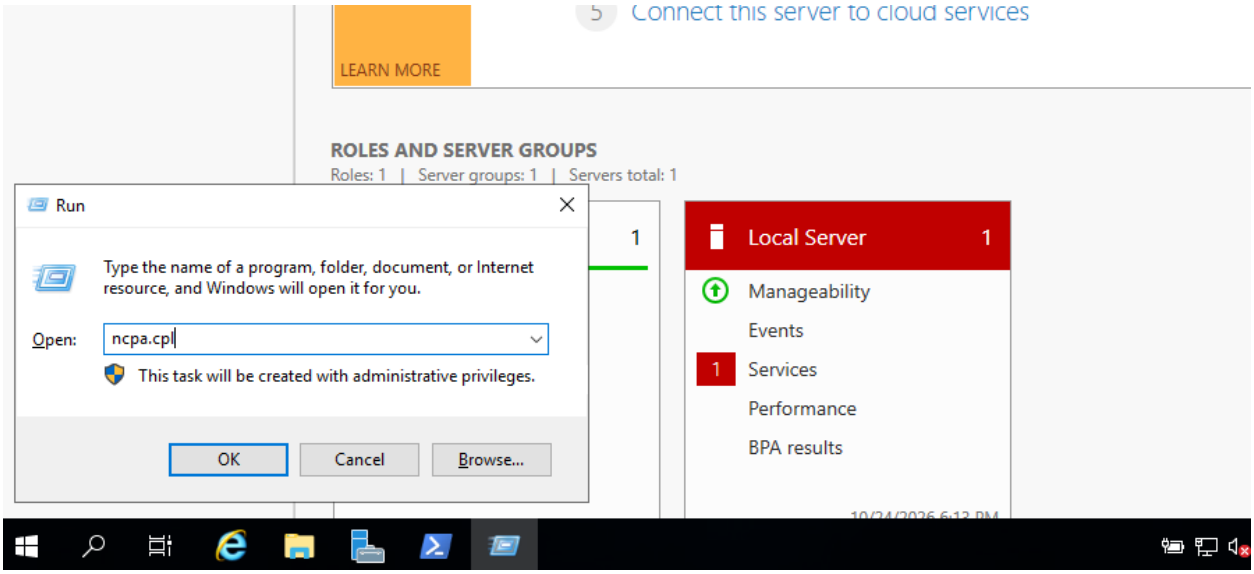


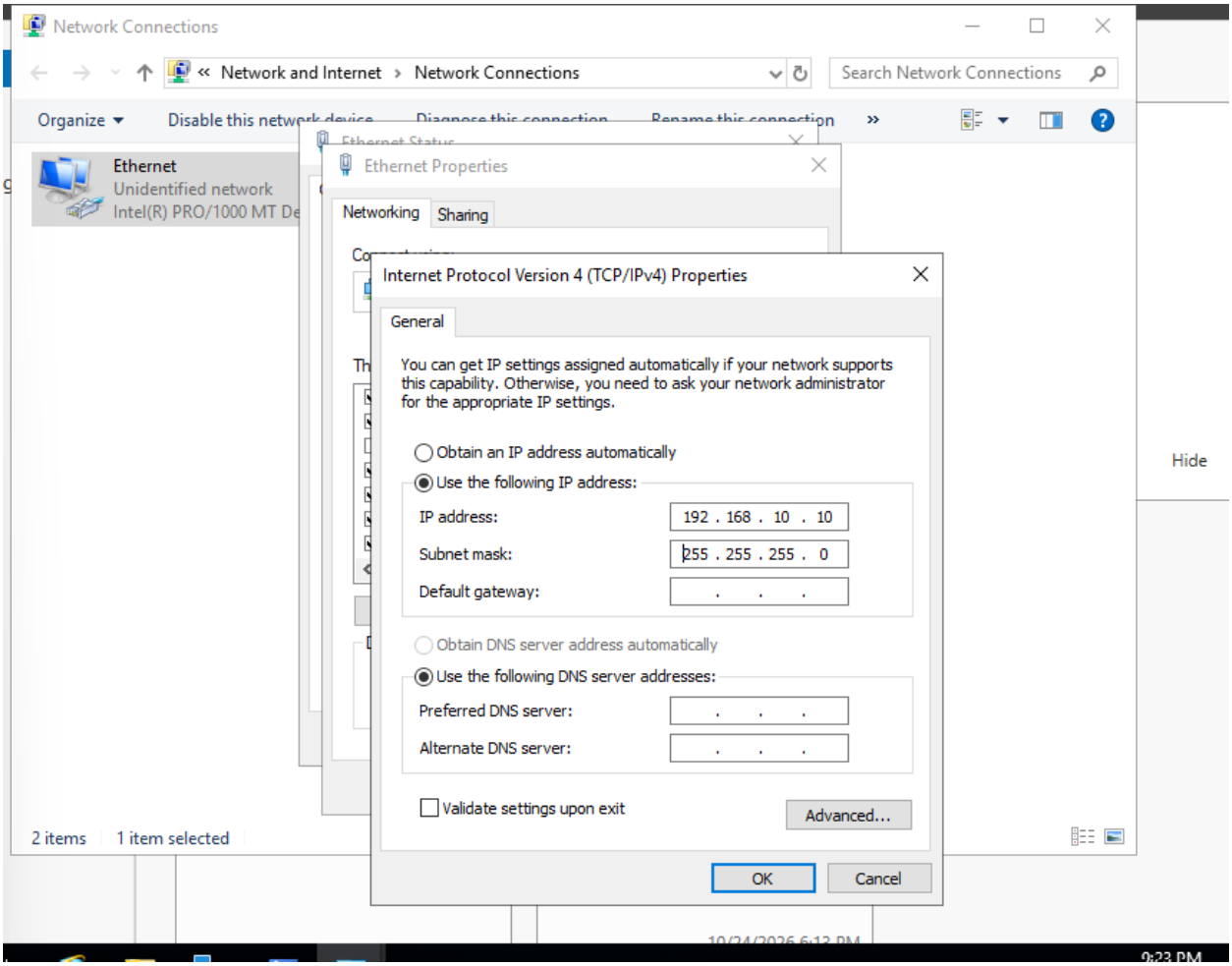


B-Vérifier la date et l'heure



C-Configurer l'adresse IPv4






```
PS C:\Users\Administrator> Get-NetAdapter

Name                      InterfaceDescription                    ifIndex Status       MacAddress             LinkSpeed
----                      --------------------                    ------- ------       ----------             ---------
Ethernet                  Intel(R) PRO/1000 MT Desktop Adapter          7 Up           08-00-27-88-F5-66         1 Gbps
Ethernet 2                Intel(R) PRO/1000 MT Desktop Adapter #2       4 Up           08-00-27-D4-D1-6F         1 Gbps


PS C:\Users\Administrator> Get-NetIPconfiguration


InterfaceAlias       : Ethernet 2
InterfaceIndex       : 4
InterfaceDescription : Intel(R) PRO/1000 MT Desktop Adapter #2
NetProfile.Name      : Network
IPv6Address          : fd17:625c:f037:3:dcdf:f81f:2ffd:d04f
IPv4Address          : 10.0.3.15
IPv6DefaultGateway   : fe80::2
IPv4DefaultGateway   : 10.0.3.2
DNSServer            : fd17:625c:f037:3::3
                       192.168.1.1

InterfaceAlias       : Ethernet
InterfaceIndex       : 7
InterfaceDescription : Intel(R) PRO/1000 MT Desktop Adapter
NetProfile.Name      : Unidentified network
IPv4Address          : 192.168.10.10
IPv6DefaultGateway   :
IPv4DefaultGateway   :
DNSServer            : fec0:0:0:ffff::1
                       fec0:0:0:ffff::2
                       fec0:0:0:ffff::3
```

```
PS C:\Users\Administrator> Get-DnsClientServerAddress -AddressFamily IPv4

InterfaceAlias               Interface Address ServerAddresses
                             Index     Family
--------------               --------- ------- --------------
Ethernet                             7 IPv4    {}
Ethernet 2                           4 IPv4    {192.168.1.1}
Loopback Pseudo-Interface 1          1 IPv4    {}
```

```
PS C:\Users\Administrator> ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : SRV-INFRA01
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : lan

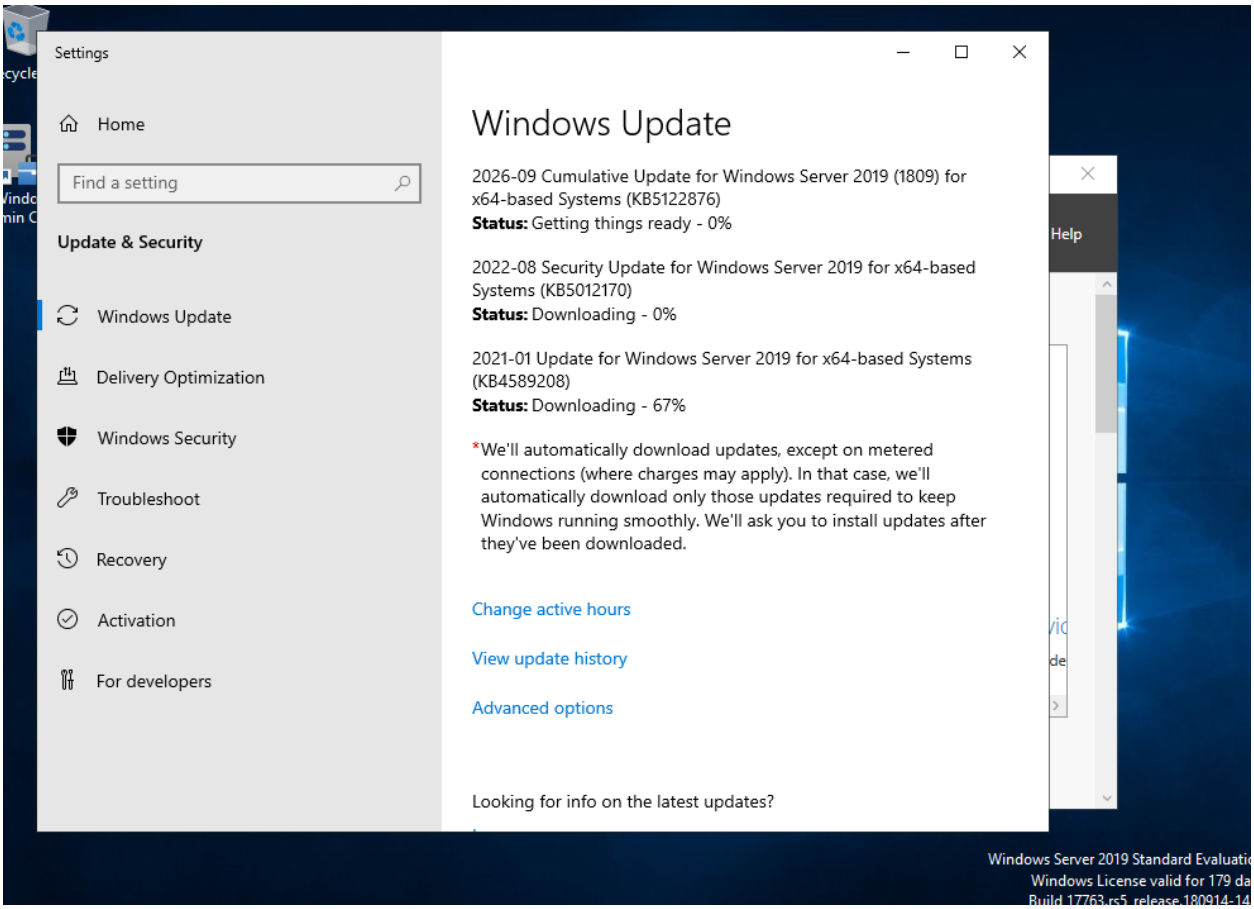
Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
   Physical Address. . . . . . . . . : 08-00-27-88-F5-66
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::cc45:6c2c:4636:cdd0%7(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.10.10(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 117964839
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-32-6F-31-81-08-00-27-88-F5-66
   DNS Servers . . . . . . . . . . . : fec0:0:0:ffff::1%1
                                       fec0:0:0:ffff::2%1
                                       fec0:0:0:ffff::3%1
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . : lan
   Description . . . . . . . . . . . : Intel(R) PRO/1000 MT Desktop Adapter #2
   Physical Address. . . . . . . . . : 08-00-27-D4-D1-6F
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : fd17:625c:f037:3:dcdf:f81f:2ffd:d04f(Preferred)
   Link-local IPv6 Address . . . . . : fe80::f22a:48c0:55ba:4f96%4(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.0.3.15(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Saturday, October 24, 2026 8:53:25 PM
   Lease Expires . . . . . . . . . . : Sunday, October 25, 2026 8:53:24 PM
   Default Gateway . . . . . . . . . : fe80::2%4
                                       10.0.3.2
   DHCP Server . . . . . . . . . . . : 10.0.3.2
   DHCPv6 IAID . . . . . . . . . . . : 67633191
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-32-6F-31-81-08-00-27-88-F5-66
```

D-Mettre le système à jour



1. Pourquoi attribuer une adresse fixe à un serveur ?

R-Pour que les clients le retrouvent toujours à la même adresse. Avec une adresse automatique, elle pourrait changer.

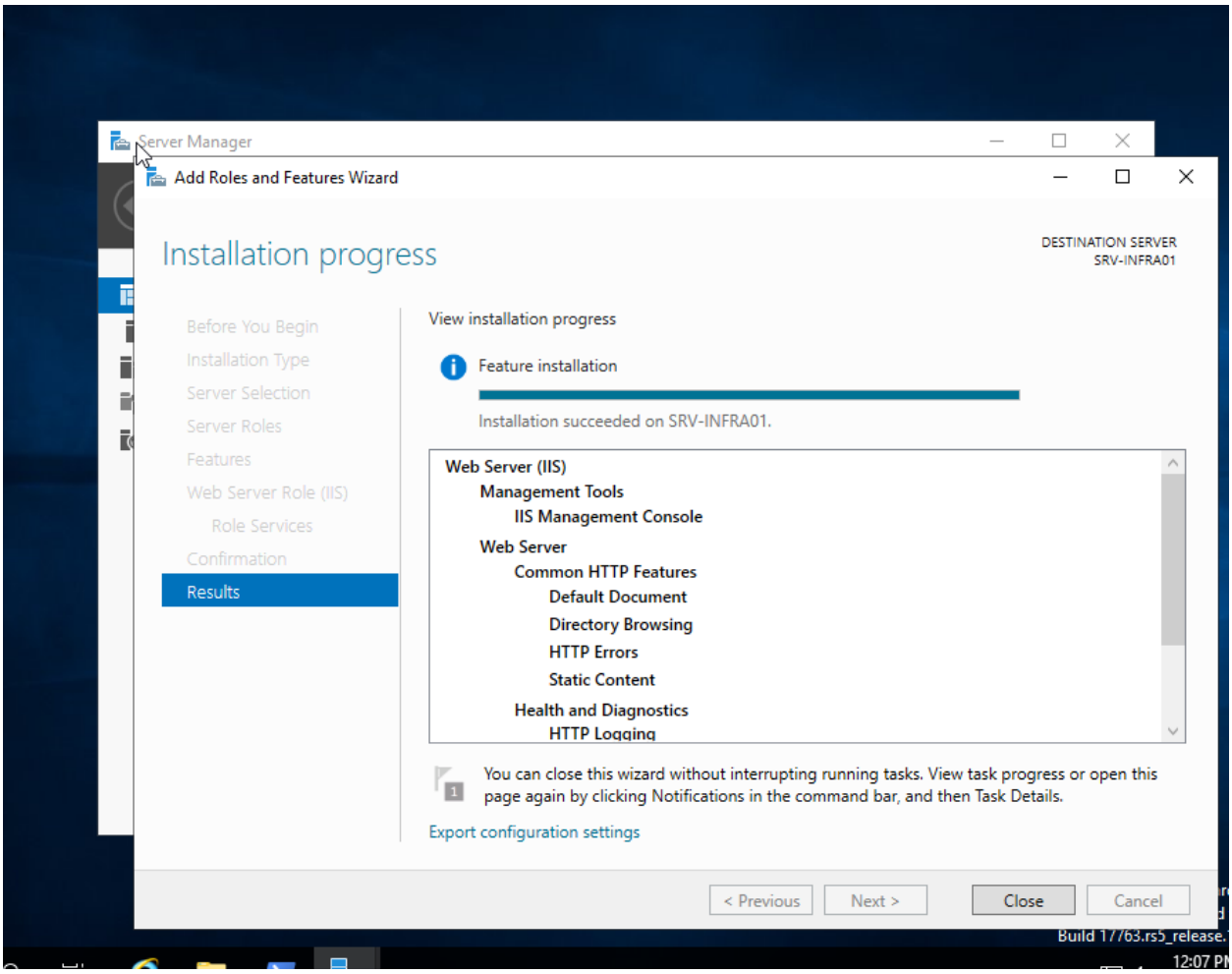
2. Quelle est la fonction du masque réseau ?

R-Il indique quelle partie de l'adresse désigne le réseau et laquelle désigne la machine. Avec `255.255.255.0`, toutes les machines `192.168.10.x` sont dans le même réseau.

3. À quoi servent la passerelle et le DNS ?

R-La passerelle est le routeur qui permet de sortir du réseau local (vers Internet). Le DNS traduit les noms (comme `SRV-INFRA01`) en adresses IP.

4-Installation du serveur Web IIS



A-Commande PowerShell équivalente

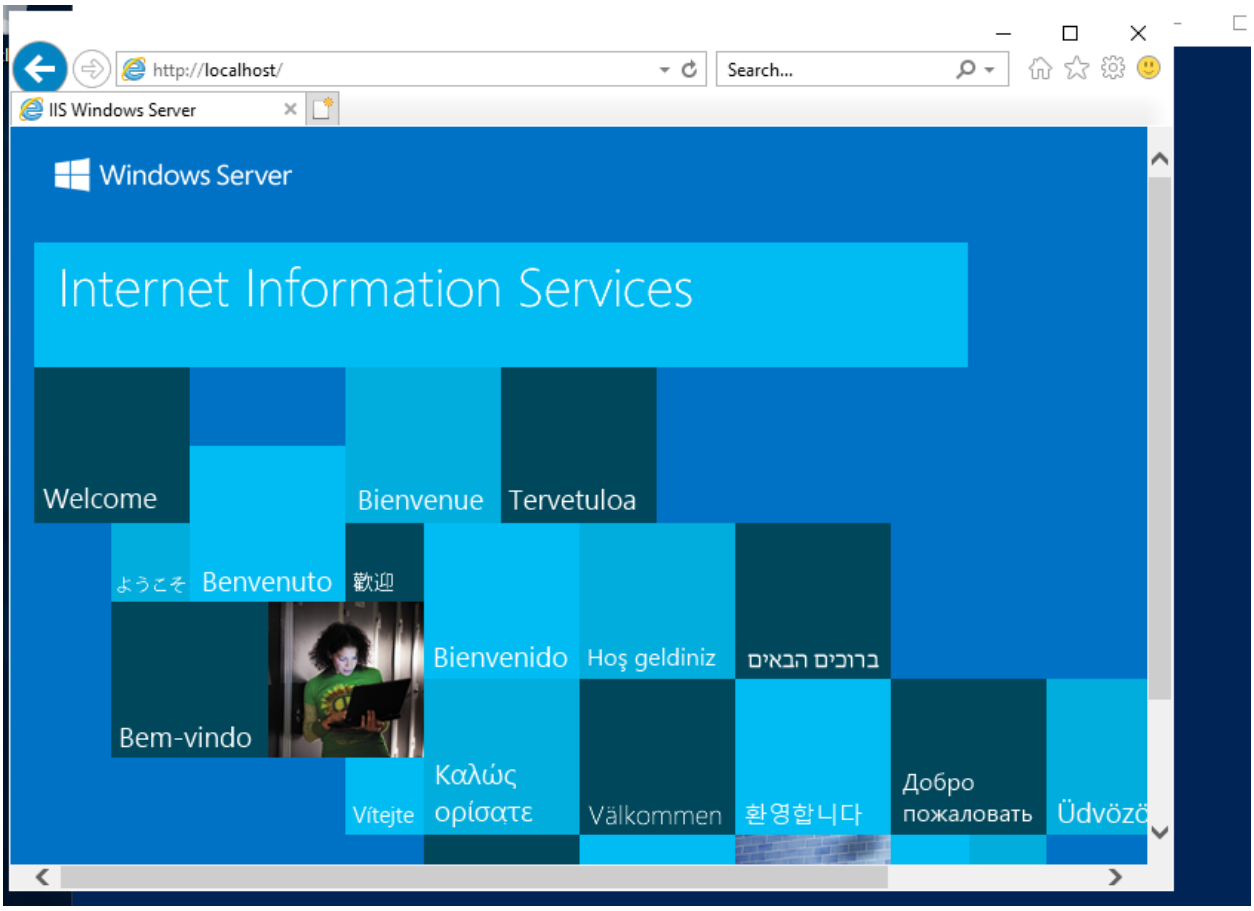
```
PS C:\Users\Administrator> Get-WindowsFeature -Name Web-Server

Display Name                                            Name                    Install State
------------                                            ----                    -------------
[X] Web Server (IIS)                                    Web-Server                  Installed


PS C:\Users\Administrator> Get-Service -Name W3SVC

Status   Name               DisplayName
------   ----               -----------
Running  W3SVC              World Wide Web Publishing Service
```

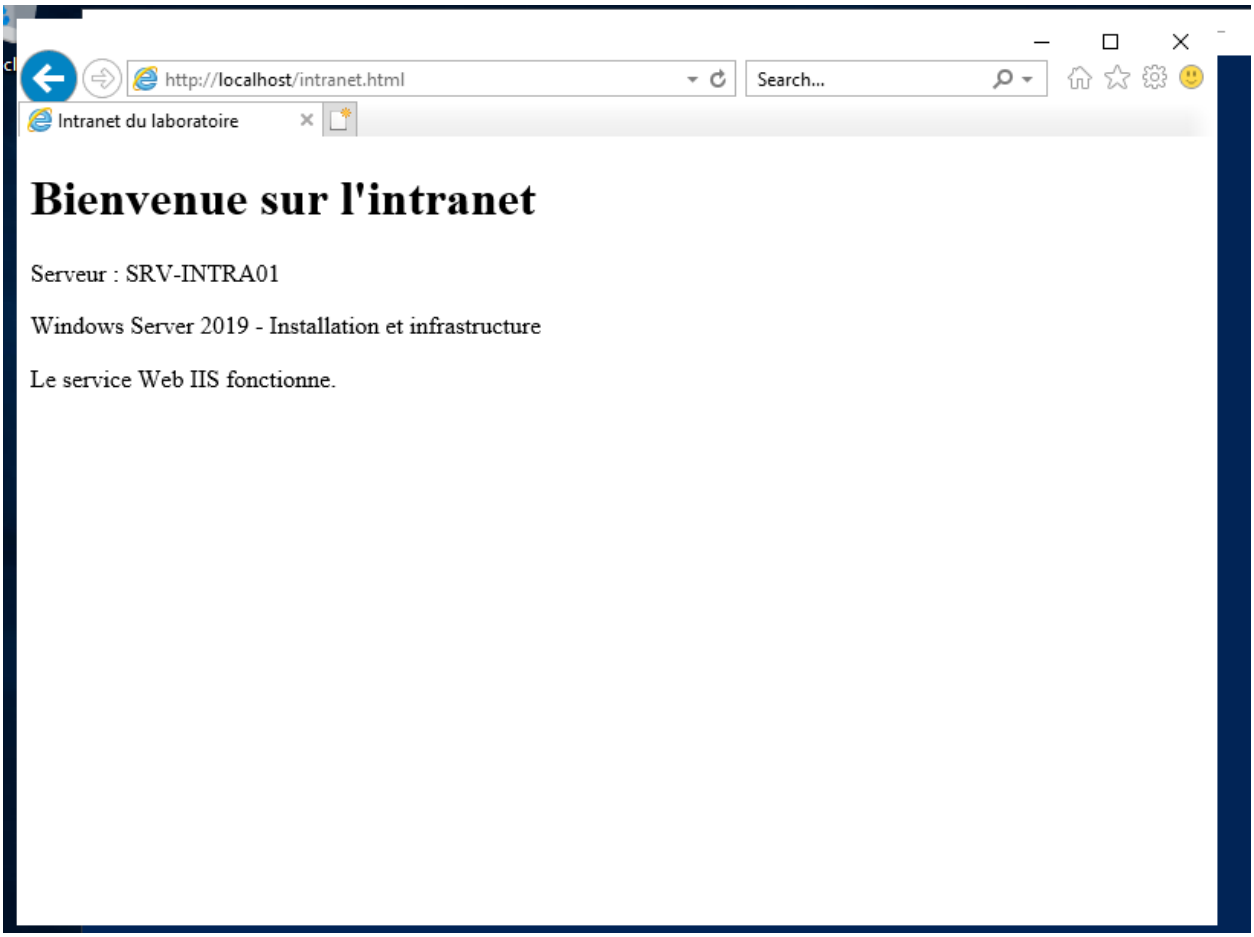
B-Vérifier le site par défaut



C-Créer une page intranet



Ouvrons




```
PS C:\Users\Administrator> Get-ChildItem C:\inetpub\wwwroot


    Directory: C:\inetpub\wwwroot


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        10/25/2026  12:04 PM            703 iisstart.htm
-a----        10/25/2026  12:04 PM          99710 iisstart.png
-a----        10/25/2026  12:46 PM            328 intranet.html


PS C:\Users\Administrator>
```

5-Installation du rôle de serveur de fichiers

```
>> '@ | Set-Content -Path "C:\inetpub\wwwroot\intranet.html" -Encoding UTF8
PS C:\Users\Administrator> Get-ChildItem C:\inetpub\wwwroot


    Directory: C:\inetpub\wwwroot


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        10/25/2026  12:04 PM            703 iisstart.htm
-a----        10/25/2026  12:04 PM          99710 iisstart.png
-a----        10/25/2026  12:46 PM            328 intranet.html


PS C:\Users\Administrator> Install-WindowsFeature -Name FS-FileServer -IncludeManagementTools

Success Restart Needed Exit Code      Feature Result
------- -------------- ---------      --------------
True    No             Success        {File and iSCSI Services, File Server}


PS C:\Users\Administrator> New-Item -Path "C:\Partages\Commun" -ItemType Directory -Force


    Directory: C:\Partages


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/25/2026  12:57 PM                Commun


PS C:\Users\Administrator> "Document de test du serveur SRV SRV-INTRA01." | Set-Content -Path "C:\Partages\Commun\Bienve
nue.text" -Encoding UTF8
PS C:\Users\Administrator>
```

```
PS C:\Users\Administrator> Get-ChildItem -Path "C:\partages\Commun"


    Directory: C:\partages\Commun


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        10/25/2026   1:02 PM             49 Bienvenue.text


C:\Users\Administrator>
```

1. Quelle différence existe-t-il entre un rôle et une fonctionnalité ?

R-Un rôle est une fonction principale du serveur (IIS, serveur de fichiers). Une fonctionnalité est un composant complémentaire (outils de gestion, sauvegarde).

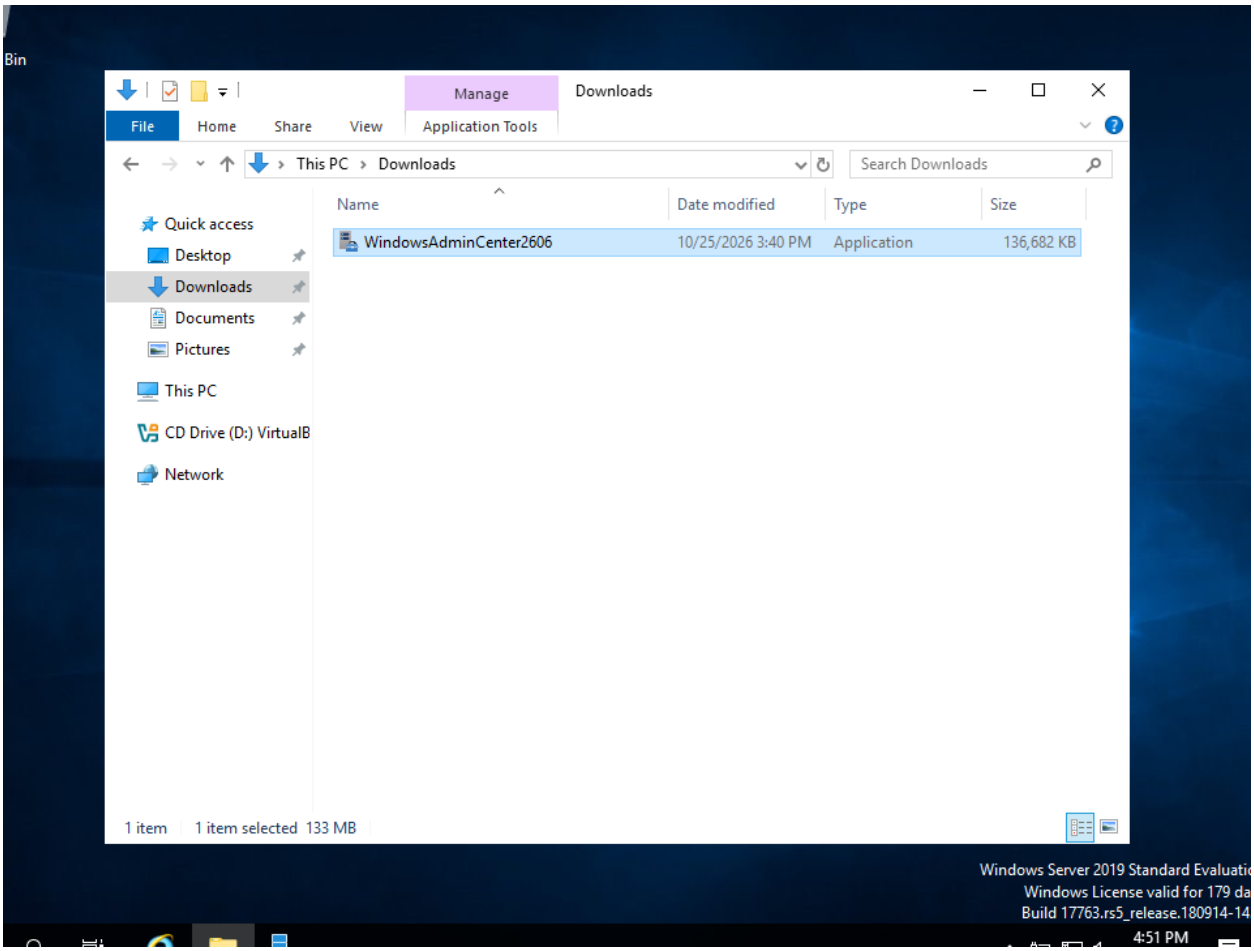
2. L'installation du rôle de serveur de fichiers crée-t-elle automatiquement un partage utilisable ?

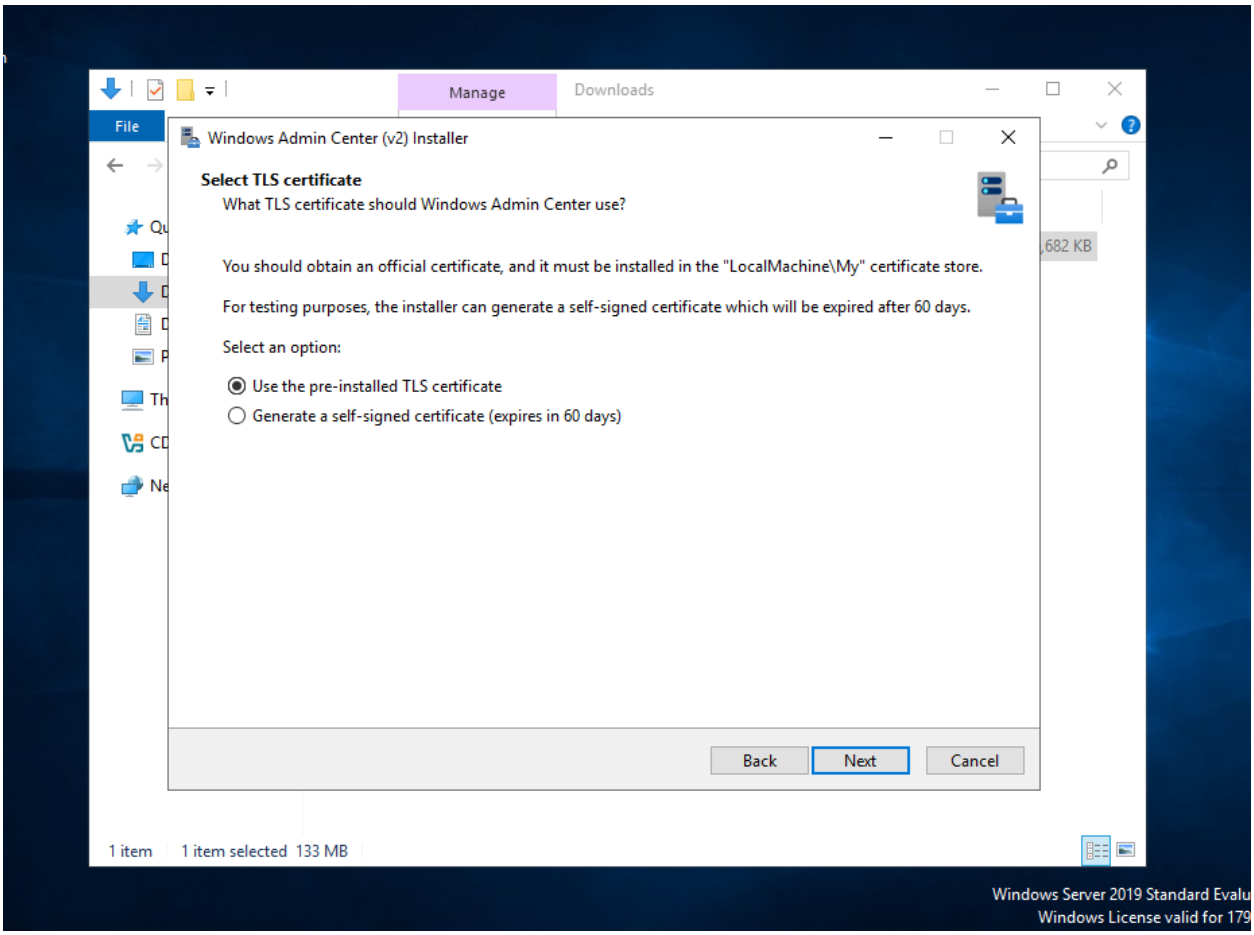
R-Non, il faut ensuite créer un partage SMB et configurer les autorisations de partage et NTFS.

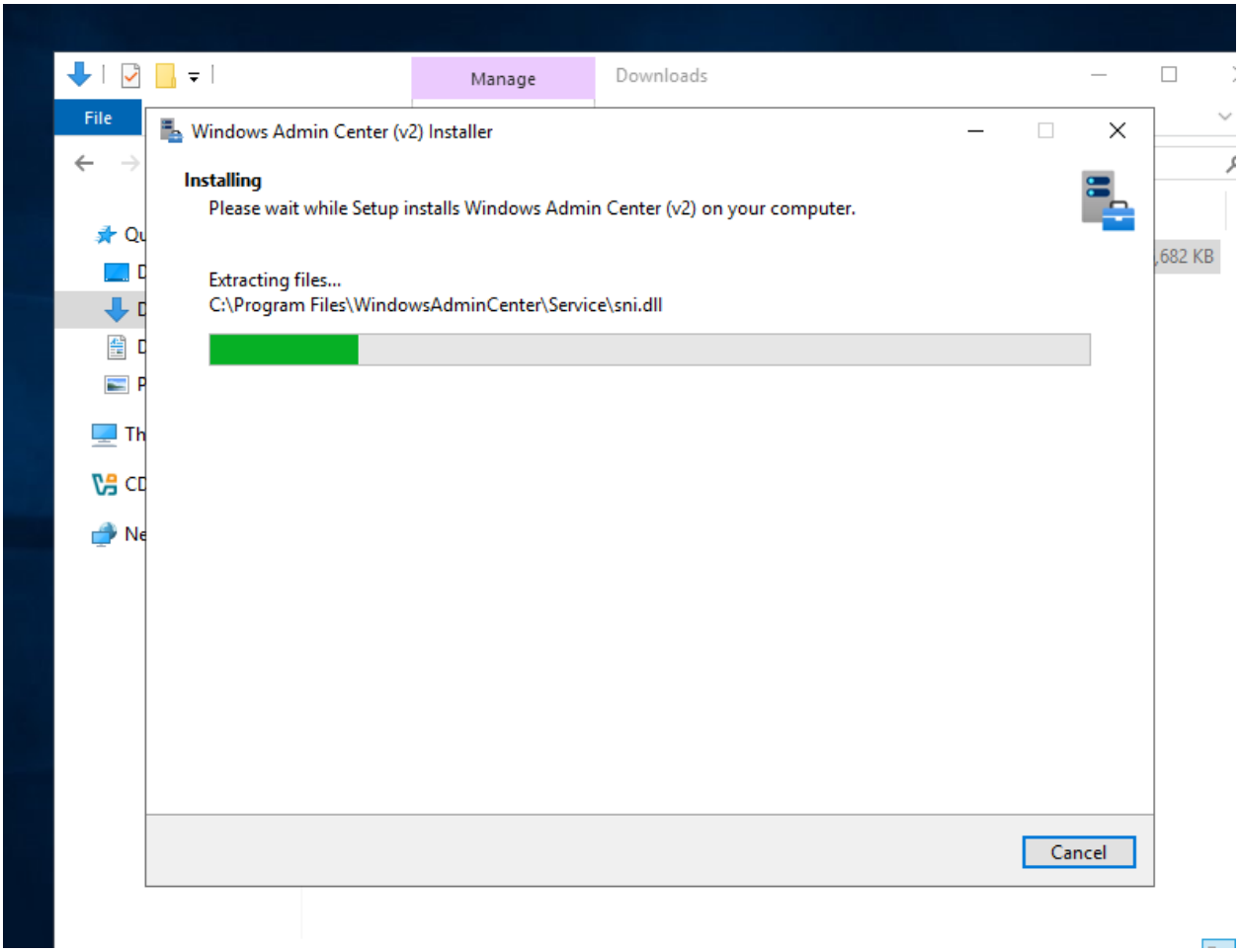
3. Quel intérêt présente PowerShell pour configurer plusieurs serveurs ?

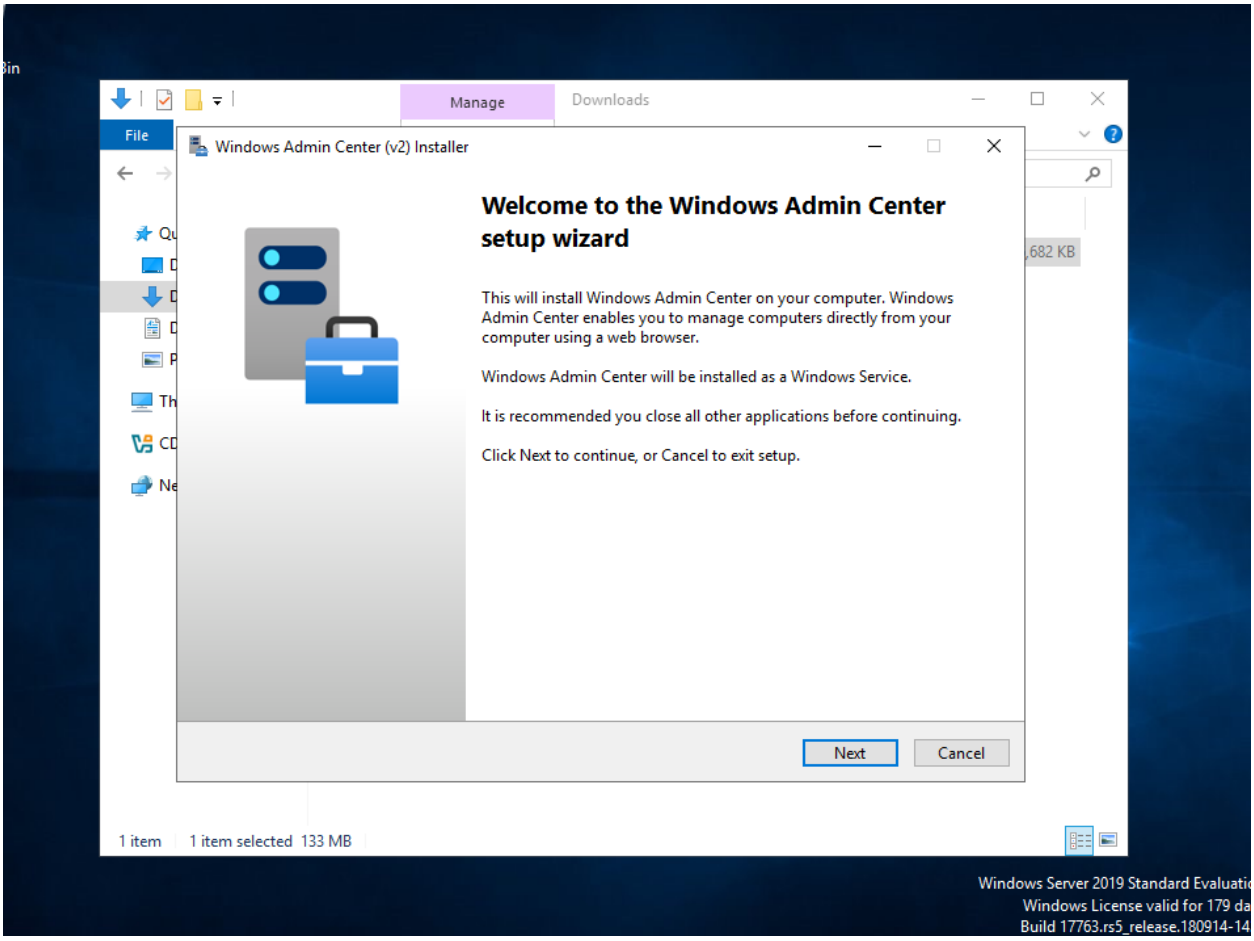
R-Il permet d'automatiser : une seule commande configure plusieurs serveurs, plus vite et sans erreur de manipulation.

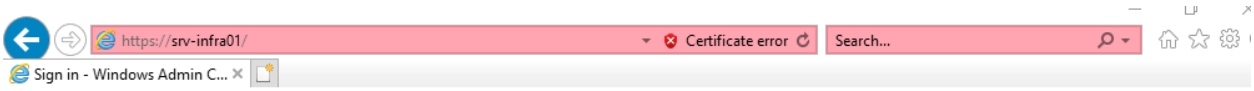
6-Installation et utilisation de Windows Admin Center

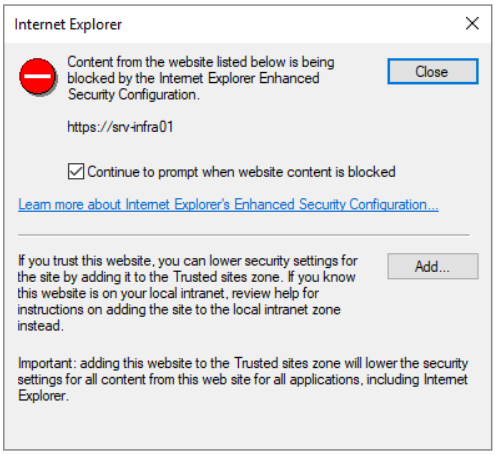


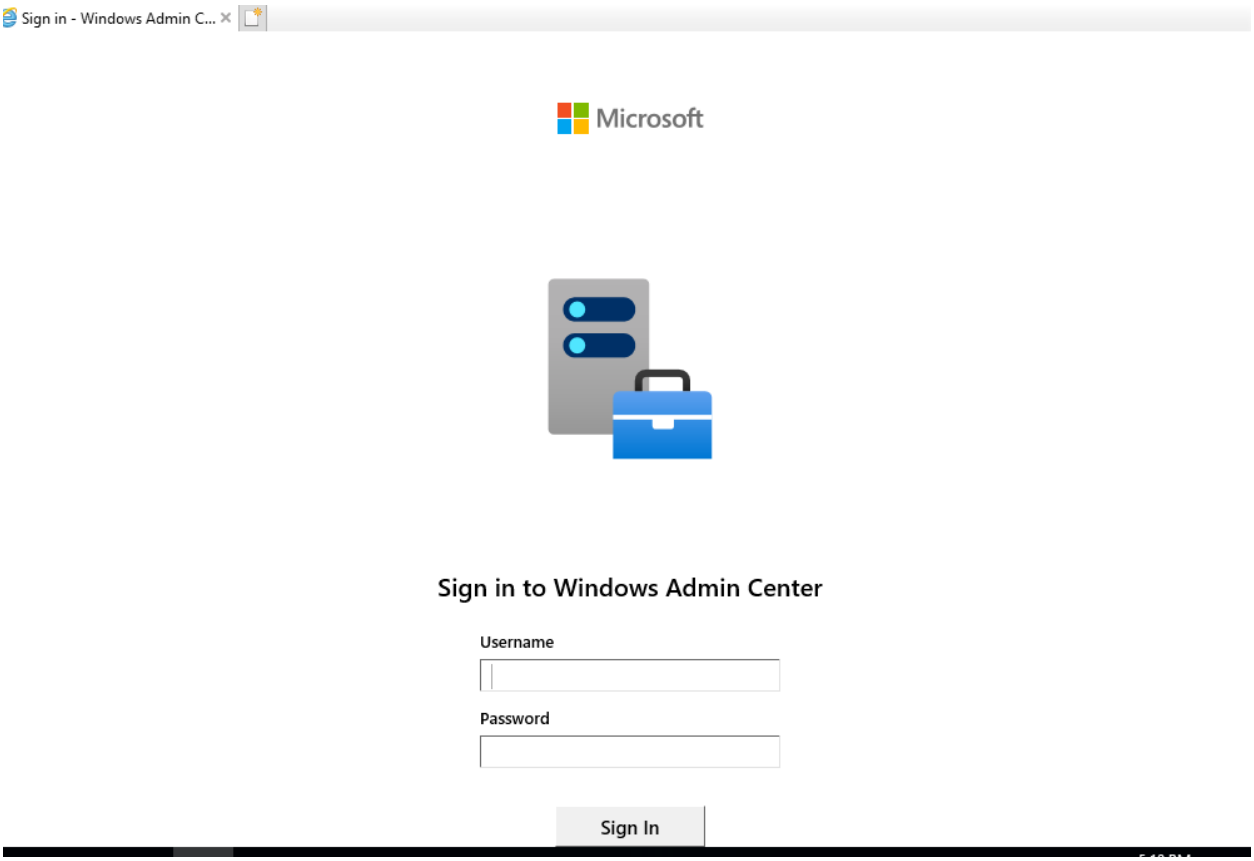

Manage
Downloads
File
Windows Admin Center (v2) Installer
Select TLS certificate
What TLS certificate should Windows Admin Center use?
You should obtain an official certificate, and it must be installed in the "LocalMachine\My" certificate store.
For testing purposes, the installer can generate a self-signed certificate which will be expired after 60 days.
Select an option:
Use the pre-installed TLS certificate
Generate a self-signed certificate (expires in 60 days)
Back
Next
Cancel
1 item
1 item selected 133 MB
Windows Server 2019 Standard Evalu
Windows License valid for 179


Manage
Downloads
File
Windows Admin Center (v2) Installer
Installing
Please wait while Setup installs Windows Admin Center (v2) on your computer.
Extracting files...
C:\Program Files\WindowsAdminCenter\Service\sni.dll
Cancel


Manage
Downloads
File
Windows Admin Center (v2) Installer
Welcome to the Windows Admin Center setup wizard
This will install Windows Admin Center on your computer. Windows Admin Center enables you to manage computers directly from your computer using a web browser.
Windows Admin Center will be installed as a Windows Service.
It is recommended you close all other applications before continuing.
Click Next to continue, or Cancel to exit setup.
Next
Cancel
1 item
1 item selected 133 MB
Windows Server 2019 Standard Evaluati
Windows License valid for 179 da
Build 17763.rs5_release.180914-14


https://srv-infra01/
Certificate error
Search...
Sign in - Windows Admin C...


Internet Explorer
Content from the website listed below is being blocked by the Internet Explorer Enhanced Security Configuration.
Close
https://srv-infra01
Continue to prompt when website content is blocked
Learn more about Internet Explorer's Enhanced Security Configuration...
If you trust this website, you can lower security settings for the site by adding it to the Trusted sites zone. If you know this website is on your local intranet, review help for instructions on adding the site to the local intranet zone instead.
Add...
Important: adding this website to the Trusted sites zone will lower the security settings for all content from this web site for all applications, including Internet Explorer.


Sign in - Windows Admin C...
Microsoft
Sign in to Windows Admin Center
Username
Password
Sign In